

Lab Report: Week 1 Session B

AI Mental Models — Chain-of-Thought Practice

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Abstract

This report documents Week 1 Session B: comparing **chain-of-thought** and **direct** prompting with OpenCode Desktop on reservoir storage (500 ha, 15 m depth $\rightarrow$ 75 MCM) and SCS-CN runoff ($P = 80$ mm, $CN = 85 \rightarrow Q = 43.6$ mm). Both runoff prompts agreed numerically; CoT required explicit verification ($Q \leq P$). Evidence includes three OpenCode screenshots and the full `prompt_log.md` in Appendix ??.

1 Introduction

This session practiced **prompting strategy** for hydrology and water-resources calculations using OpenCode Desktop with OpenCode Zen. Two styles were compared: a **chain-of-thought (CoT)** prompt that requests numbered reasoning steps, and a **direct** one-line question. Exercises covered reservoir storage volume (unit conversion and magnitude check) and SCS-CN direct runoff for 80 mm rainfall with curve number 85.

2 Objectives

- Practice CoT prompting with explicit steps and verification.
- Compare direct vs. step-by-step AI responses on the same hydrologic problem.
- Document prompts, outputs, and verification in a prompt log.
- Reflect on when each prompting style is appropriate.

3 Methods

Work was carried out in `ai_water_lab/week1_session_b/` using OpenCode Desktop. No Python script was required for this lab; results are AI-generated calculations recorded in `prompt_log.md` and supported by OpenCode session screenshots (Figures ??–??).

3.1 Exercise 1: Reservoir capacity (CoT)

The lab prompt specified 500 ha surface area and 15 m average depth, with steps to convert hectares to square metres, compute volume, express result in million cubic metres (MCM), and comment on whether the value is reasonable for a medium-sized reservoir.

3.2 Exercise 2: SCS-CN runoff (direct vs. CoT)

The same rainfall depth ($P = 80$ mm) and curve number ($CN = 85$) were used twice:

- **Direct:** *Calculate the runoff for 80mm rainfall with $CN=85$*

- **CoT**: same problem with explicit steps for S , I_a , the SCS-CN formula, and verification $Q \leq P$.

The standard SCS-CN relations used were $S = (25400/CN) - 254$, $I_a = 0.2S$, and for $P > I_a$, $Q = (P - I_a)^2 / (P - I_a + S)$.

4 Results

4.1 Exercise 1: Reservoir storage

The AI solution gave:

- Area: 500 ha = 5,000,000 m²
- Volume: 75,000,000 m³
- Storage: **75 MCM**

Figure ?? shows the OpenCode conversation for this prompt.

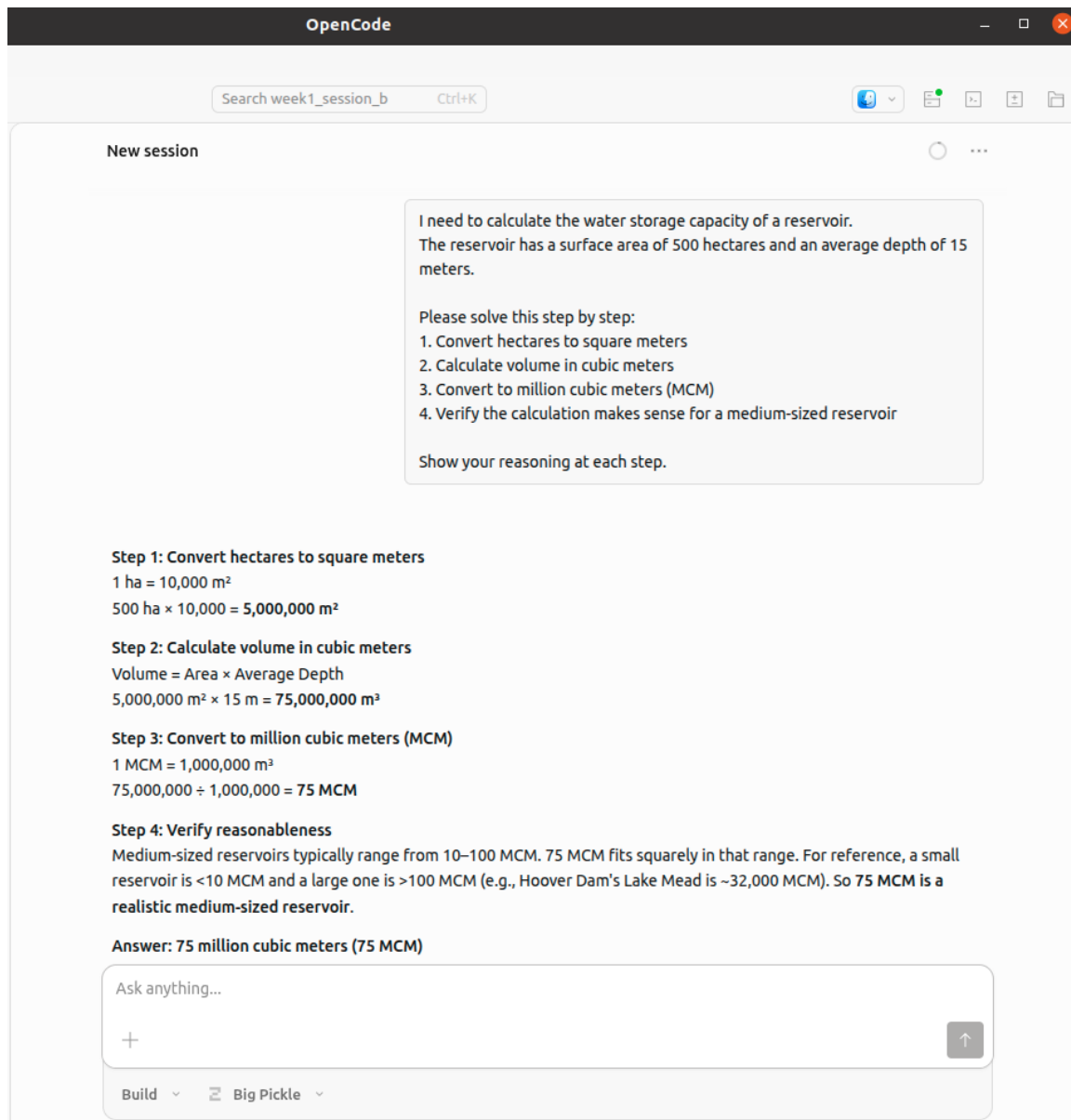


Figure 1: OpenCode session: chain-of-thought reservoir capacity calculation (Exercise 1).

4.2 Exercise 2: Direct SCS-CN prompt

The direct prompt produced $S = 44.82$ mm, $I_a = 8.96$ mm, and runoff $Q = 43.6$ mm, with intermediate algebra shown. Figure ?? documents this exchange.

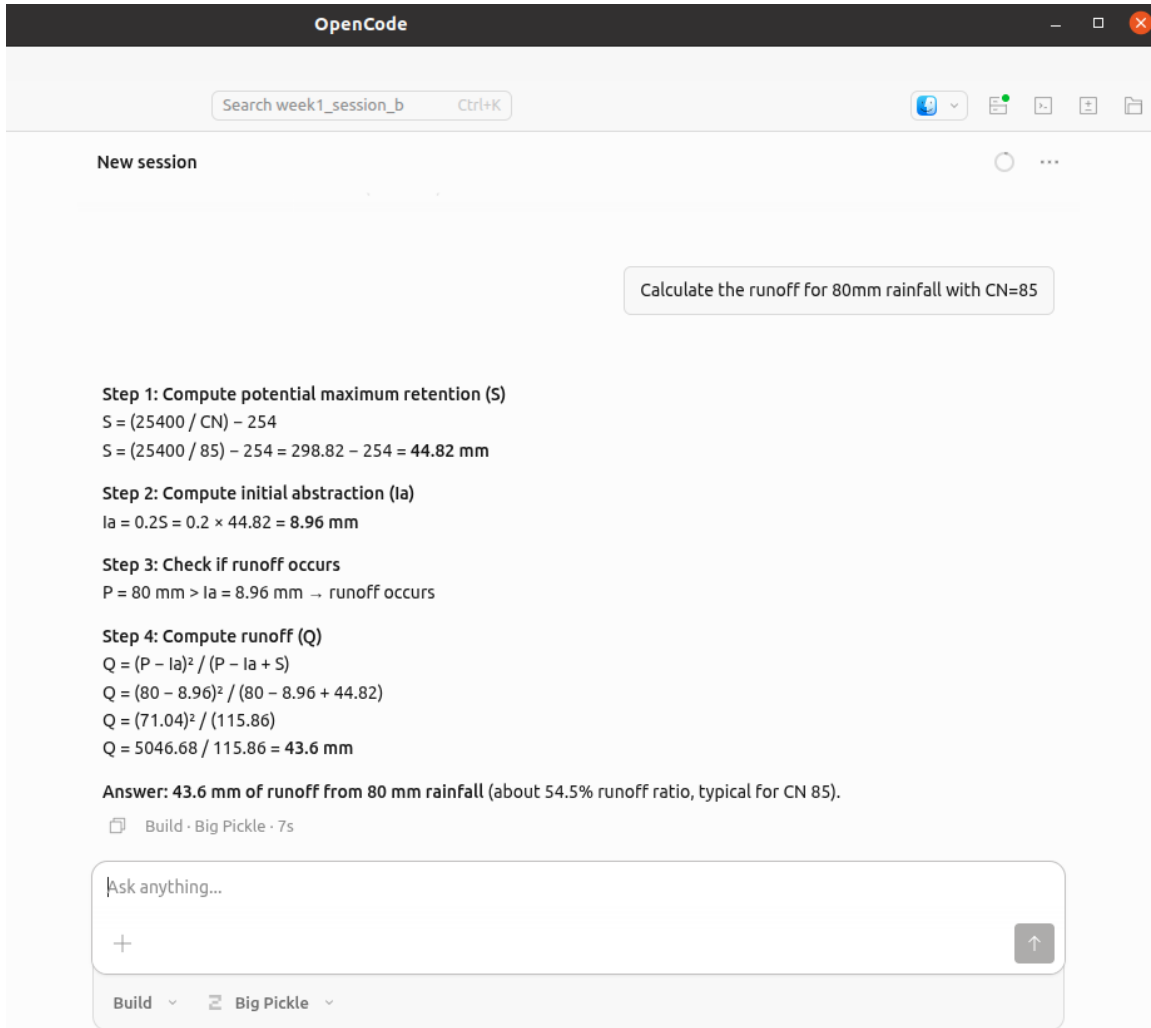


Figure 2: OpenCode session: direct prompt for SCS-CN runoff ($P = 80$ mm, $CN = 85$).

4.3 Exercise 2: CoT SCS-CN prompt

The structured prompt yielded the same $Q = 43.6$ mm and explicitly verified $Q \leq P$. Figure ?? shows this session.

Table ?? summarizes both runoff prompts.

Table 1: Comparison of direct vs. CoT prompts for SCS-CN runoff ($P = 80$ mm, $CN = 85$).

Item	Direct prompt	CoT prompt
Final Q	43.6 mm	43.6 mm
Steps shown	Yes (unprompted)	Yes (requested)
$Q \leq P$ check	Implicit in narrative	Explicit step

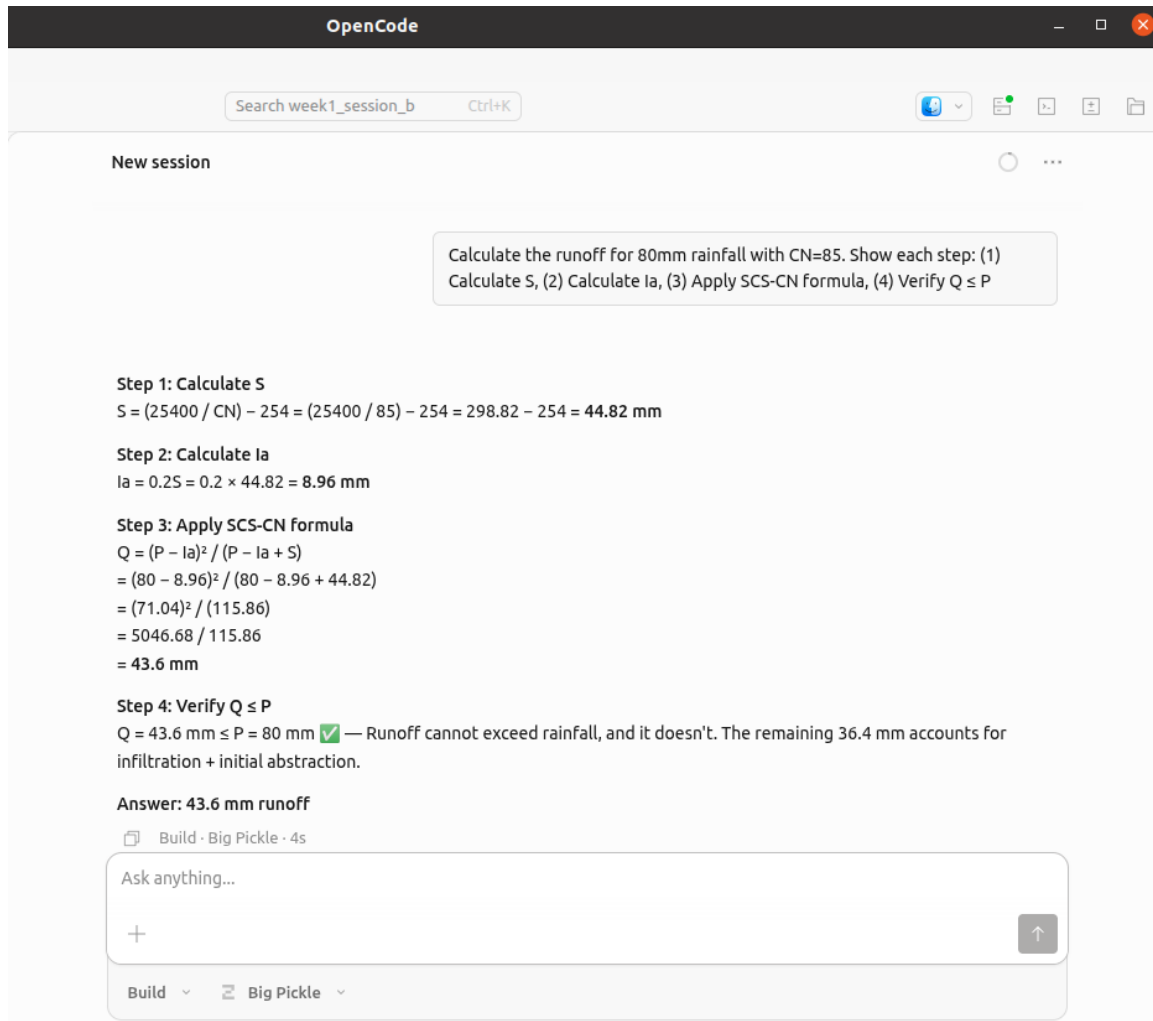


Figure 3: OpenCode session: chain-of-thought SCS-CN calculation with $Q \leq P$ verification.

5 Analysis and validation

Reservoir (Exercise 1): Manual recomputation confirms 75 MCM. The CoT structure separated unit conversion, volume, and scaling, which reduces the risk of mixing hectares with square metres or forgetting the 10^6 factor for MCM.

SCS-CN (Exercise 2): Independent check: $S = 25400/85 - 254 = 44.82$ mm; $I_a = 8.96$ mm; $P - I_a = 71.04$ mm; $Q = 71.04^2/115.86 \approx 43.6$ mm; $Q < P$. Both AI answers match this check.

Direct vs. CoT: The numeric answers agreed. The main difference is *process control*: CoT prompts force verification steps that are easy to grade and to defend in a report. The direct prompt still produced detailed work here, but that behaviour is not guaranteed for every model or every problem.

Limitations: AI reasoning is not a substitute for design standards, antecedent moisture adjustments, or site data. Reservoir “reasonableness” ranges cited by the model are qualitative.

6 Discussion

1. CoT increased confidence because each step could be checked against textbook formulas; the direct answer was equally numeric but less structured.
2. Step-by-step prompts suit multi-unit conversions, method-based hydrology (SCS-CN, routing), and any graded work requiring shown work.
3. Even with “show your work,” the user must verify units, boundary conditions ($P < I_a$), and $Q \leq P$.
4. Next session: specify output format and verification checks in the prompt from the start (as in the CoT runoff prompt).

7 Problems encountered

No calculation errors were found after manual verification. The main practical issue is documenting three separate OpenCode exchanges; screenshots were saved for each prompt.

8 Lessons learned

Structured prompts align AI output with engineering documentation habits. When direct and CoT answers match, CoT still adds audit value. Keeping a prompt log alongside screenshots makes report writing straightforward and satisfies the course submission requirements.

9 Conclusion

Week 1 Session B objectives were met: CoT reservoir calculation (75 MCM), comparison of direct and CoT SCS-CN runoff (both 43.6 mm), and documentation in `prompt_log.md` with OpenCode evidence. Course submission: `prompt_log.md` and screenshots as specified in the lab guide.

A Submitted prompt log: prompt_log.md

prompt_log.md (full file)

Prompt Log - Week 1 Session B

Exercise 1: Reservoir storage (CoT)

Date: 2026-05-16

Prompt used:

I need to calculate the water storage capacity of a reservoir.
The reservoir has a surface area of 500 hectares and an average depth of 15 meters.

Please solve this step by step:

1. Convert hectares to square meters
2. Calculate volume in cubic meters
3. Convert to million cubic meters (MCM)
4. Verify the calculation makes sense for a medium-sized reservoir

Show your reasoning at each step.

AI output summary:

OpenCode returned a four-step chain-of-thought solution: (1) 500 ha = 5,000,000 m using 1
→ ha = 10,000 m; (2) volume = 5,000,000 m 15 m = 75,000,000 m; (3) 75 MCM; (4) sanity
→ check that 75 MCM lies in a typical medium-reservoir range (about 10100 MCM), with
→ brief comparison to smaller and very large impoundments.

Verification:

- Unit chain: hectares → m² → m³ → MCM is consistent (MCM = m³ / 10).
- Arithmetic: 500 × 10,000 = 5,000,000; × 15 = 75,000,000 m³; ÷ 10 = 75 MCM.
- Magnitude: 75 MCM is plausible for a medium-sized storage project; the AIs qualitative
→ check is reasonable for coursework (not a substitute for site-specific design data).

Lessons learned:

Numbered steps in the prompt produced a response that was easy to audit step-by-step.
→ Explicit verify reasonableness encouraged a short engineering judgment, not only
→ algebra.

Exercise 2: Direct vs CoT (SCS-CN)

Direct prompt:

Calculate the runoff for 80mm rainfall with CN=85

Direct response summary:

Despite the short prompt, the model still showed intermediate steps: S = 44.82 mm, Ia =
→ 8.96 mm, check P > Ia, then Q = 43.6 mm using the SCS-CN formula, plus a runoff-ratio
→ comment (~54.5%).

CoT prompt:

Calculate the runoff for 80mm rainfall with CN=85. Show each step: (1) Calculate S, (2)
→ Calculate Ia, (3) Apply SCS-CN formula, (4) Verify Q P

CoT response summary:

Same intermediate values ($S = 44.82$ mm, $I_a = 8.96$ mm) and final $Q = 43.6$ mm. Step 4
↪ explicitly verified $Q - P$ (43.6 mm $- 80$ mm) and interpreted the remainder as
↪ infiltration plus initial abstraction.

****Comparison:****

- ****Numeric result:**** Both prompts gave **$Q = 43.6$ mm** (consistent with manual check).
- ****Transparency:**** Direct prompt still showed work in this case; CoT prompt ****required****
↪ the audit step $Q - P$, which is valuable for grading and for catching impossible
↪ results.
- ****When to use each:**** Use a ****direct**** prompt for quick checks when you already know
↪ the method. Use ****CoT**** for coursework, design documentation, and any problem where
↪ you must show verification steps to an instructor or reviewer.